

Problem Set 2: Solution

1. Sinusoid through an LTI system

An LTI system has $H(e^{j\omega}) = 1 + 0.5e^{-j\omega}$. The input is $x[n] = 2\cos(\pi n/3 + \pi/4)$.

- Evaluate $H(e^{j\pi/3})$ in rectangular form.
- Find its magnitude and phase.
- Write the steady-state output $y[n]$ in cosine form.

$$H(e^{j\omega}) = 1 + 0.5 e^{-j\omega}$$

$$= 1 + 0.5 \cos \omega - j 0.5 \sin \omega$$

$$\begin{aligned} \text{a) } H(e^{j\pi/3}) &= 1 + 0.5 \underbrace{\cos \pi/3}_{= 1/2} - j 0.5 \underbrace{\sin \pi/3}_{= \sqrt{3}/2} \\ &= \frac{5}{4} - j \frac{\sqrt{3}}{4} = 1.25 - j 0.433 \end{aligned}$$

$$\text{b) } |H(e^{j\pi/3})| = \frac{\sqrt{25 + 3}}{4} = \frac{\sqrt{28}}{4} = 1.3229$$

$$\angle H(e^{j\pi/3}) = -\arctan\left(\frac{\sqrt{3}}{5}\right) = -0.3335 \text{ rad}$$

$$= -19.1^\circ$$

$$\begin{aligned} \text{c) } y[n] &= 2 |H(e^{j\pi/3})| \cos\left(\frac{\pi n}{3} + \frac{\pi}{4} + \angle H(e^{j\pi/3})\right) \\ &= 2.646 \cdot \cos(1.047n + 0.4519) \end{aligned}$$

2. DTFT shifting and a spectral zero

Let $x[n] = (0.5)^n u[n]$ and $y[n] = x[n-2] - x[n]$.

a) Find $X(e^{j\omega})$.

b) Use the time-shift property to find $Y(e^{j\omega})$.

c) Evaluate $Y(e^{j0})$ and explain what that value says about the sum of $y[n]$.

$$\begin{aligned} \text{a) } X(e^{j\omega}) &= \sum_n x[n] e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} = \frac{1}{1 - \frac{1}{2} e^{-j\omega}} \end{aligned}$$

$$\text{b) } Y(e^{j\omega}) = (e^{-j2\omega} - 1) X(e^{j\omega}) = \frac{e^{-j2\omega} - 1}{1 - \frac{1}{2} e^{-j\omega}}$$

$$\text{c) } Y(e^{j0}) = \frac{1 - 1}{1 - \frac{1}{2}} = 0$$

$$Y(e^{j\omega}) = \sum_n y[n] e^{-j\omega n}$$

$$\Rightarrow Y(e^{j0}) = \sum_n y[n]$$

Since $Y(e^{j0}) = 0 \Rightarrow$ The samples of the sequence $y[n]$ sum to 0.

3. Difference equation to frequency response

A causal system satisfies $y[n] - 0.8y[n-1] = x[n] - x[n-1]$, with zero initial conditions.

a) Derive $H(e^{j\omega})$.

b) Evaluate H at $\omega = 0$ and $\omega = \pi$.

c) Based only on those values and the structure of the numerator, is the system more "low-pass-like" or "high-pass-like"? Explain.

d) Use `scipy.signal.freqz` to check your reasoning.

$$a) x[n] = e^{j\omega n} \Rightarrow y[n] = H(e^{j\omega}) e^{j\omega n}$$

$$y[n] - 0.8y[n-1] = x[n] - x[n-1]$$

$$\downarrow$$
$$H(e^{j\omega}) e^{j\omega n} (1 - 0.8 e^{-j\omega}) = e^{j\omega n} (1 - e^{-j\omega})$$

$$\Rightarrow H(e^{j\omega}) = \frac{1 - e^{-j\omega}}{1 - 0.8 e^{-j\omega}}$$

$$b) H(e^{j\omega})|_{\omega=0} = \frac{1-1}{1-0.8} = 0$$

$$H(e^{j\omega})|_{\omega=\pi} = \frac{2}{1.8} \approx 1.111$$

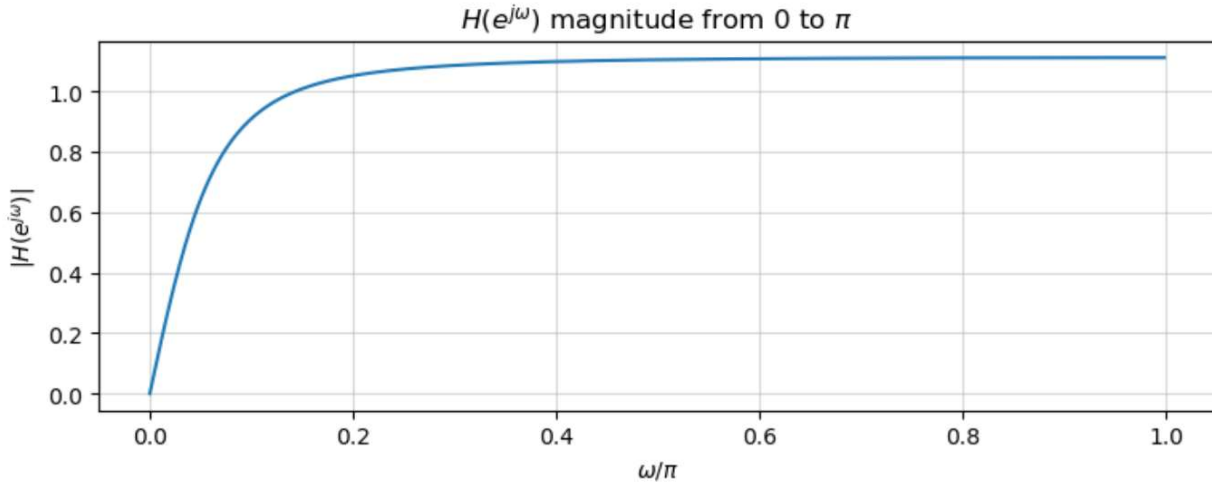
c) More "high-pass-like" since $H(e^{j\omega}) = 0$ at $\omega=0$ (dc) and $|H(e^{j\omega})| > 0$ at $\omega=\pi$ (highest normalized frequency).

Use `scipy.signal.freqz` to check your reasoning

```
import numpy as np
import scipy.signal as ss
import matplotlib.pyplot as plt
```

```
# Compute frequency response of  $H(e^{j\omega})$ 
b = [1, -1]      # numerator of  $H(e^{j\omega})$ 
a = [1, -0.8]    # denominator of  $H(e^{j\omega})$ 
[w, Hejw] = ss.freqz(b, a)  # frequency response
```

```
fig = plt.figure(figsize=(9, 3))
plt.plot(w/np.pi, np.abs(Hejw))
plt.xlabel(r'$\omega/\pi$')
plt.ylabel(r'$|H(e^{j\omega})|$')
plt.title(r'$H(e^{j\omega})$ magnitude from 0 to $\pi$')
plt.grid(True, alpha=0.5)
plt.show()
```



4. From continuous frequency to discrete frequency

A microphone voltage is $x_c(t) = 2 \cos(2\pi \cdot 1000 t) + \cos(2\pi \cdot 5000 t)$. It is sampled at $f_s = 16 \text{ kHz}$.

- a) Write $x[n]$ explicitly.
- b) Give the two discrete-time angular frequencies in rad/sample and as multiples of π .
- c) Are either components aliased at this sample rate? Explain.

$$\begin{aligned} \text{a) } x[n] &= x_c(nT) = x_c\left(\frac{n}{f_s}\right) \\ &= 2 \cos\left(2\pi \frac{1000 n}{16000}\right) + \cos\left(2\pi \frac{5000 n}{16000}\right) \\ &= 2 \cos\left(\frac{\pi n}{8}\right) + \cos\left(\frac{5\pi n}{8}\right) \end{aligned}$$

$$\begin{aligned} \text{b) } \omega_1 &= \frac{\pi}{8} \text{ rad/sample} = 0.125\pi \\ \omega_2 &= \frac{5\pi}{8} \text{ rad/sample} = 0.625\pi \end{aligned}$$

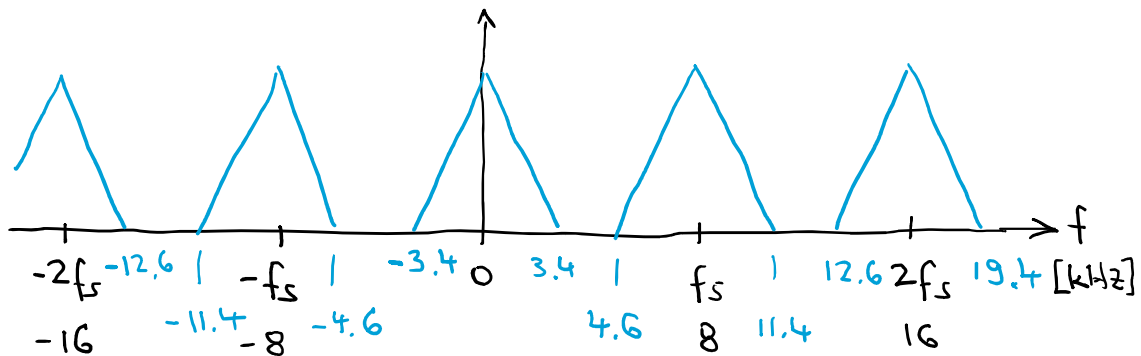
c) Neither component is aliased because both CT frequencies (1000 Hz and 5000 Hz) are below $f_s/2 = 8000 \text{ Hz}$. Equivalently, the normalized DT frequencies ω_1, ω_2 lie in $[0, \pi]$

5. Spectral replicas and Nyquist margin

An analog signal is strictly bandlimited to $|f| \leq 3.4$ kHz and is sampled at $f_s = 8$ kHz.

- Sketch or describe the locations of the baseband spectrum and the first two pairs of replicas after ideal sampling.
- What is the separation between adjacent spectral replicas?
- What guard-band width exists between the positive edge of the baseband copy and the nearest edge of the $+f_s$ replica?
- What is the ideal Nyquist rate for this signal? Is 8 kHz above it?

a) Sketch of spectrum from $-2f_s$ to $2f_s$



b) Separation between spectral replicas
is $f_s = 8$ kHz

c) Guard-band width: $(f_s - f_m) - f_m =$
 $= 8 - 2 \times 3.4 = 1.2$ kHz, $f_m = 3.4$ kHz

d) Ideal Nyquist rate $= 2 \times f_m = 6.8$ kHz < 8 kHz

6. Downsampling audio: Filter first

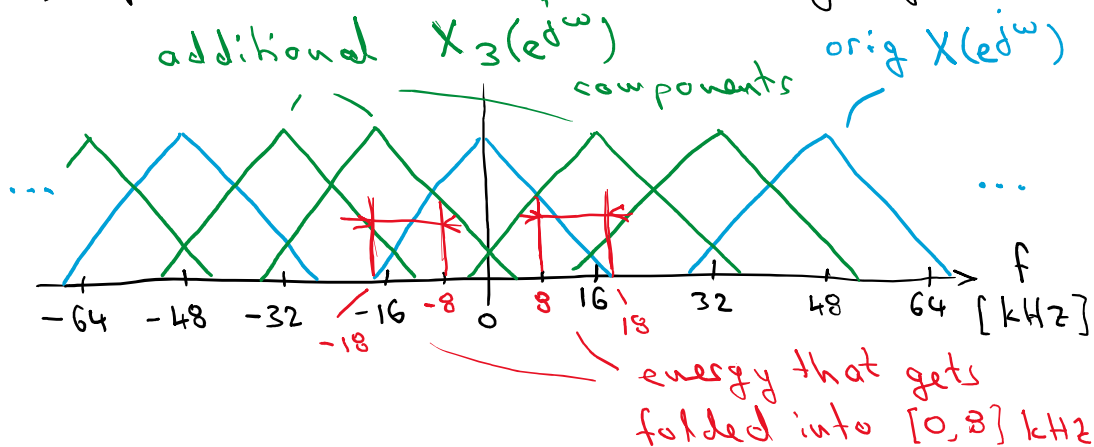
A recording is sampled at 48 kHz and contains useful content up to 18 kHz. You want a 16 kHz output by downsampling by 3.

- What is the new Nyquist frequency?
- Why is keeping every third sample without filtering incorrect?
- What band must be strongly attenuated before downsampling?
- If an 11 kHz sinusoid is present and you downsample naively to 16 kHz, at what frequency does it appear?

a) $f_{s1} = 48 \text{ kHz}$, $f_M = 18 \text{ kHz}$, $f_{s2} = \frac{f_{s1}}{3} = 16 \text{ kHz}$

New Nyquist frequency is $f_{s2}/2 = 8 \text{ kHz}$

b) Spectrum after downsampling by 3



The original signal has energy in 8 to 18 kHz band that gets folded into 0 to 8 kHz by sample reduction without filtering.

- The band that must be strongly attenuated before downsampling is 8 to 18 kHz.
- $11 \text{ kHz} = 16 \text{ kHz} - 5 \text{ kHz} \Rightarrow$ naive down-sampling by 3 would result in 5 kHz.

7. Prove two analog sinusoids give the same samples (ECEN 5632 problem)

At $f_s = 12$ kHz, compare $x_1(t) = \cos(2\pi \cdot 2000t)$ and $x_2(t) = \cos(2\pi \cdot 10000t)$.

- a) Write $x_1[n]$ and $x_2[n]$.
- b) Use a trigonometric or complex-exponential identity to prove $x_1[n] = x_2[n]$ for every integer n .
- c) Explain in one sentence why this proves aliasing cannot generally be undone after sampling.

$$\begin{aligned} \text{a) } x_1[n] &= x_{1c}(nT) & T &= \frac{1}{f_s} = \frac{1}{12000} \\ &= \cos\left(2\pi \frac{2000n}{12000}\right) = \cos\left(\frac{\pi n}{3}\right) \end{aligned}$$

$$\begin{aligned} x_2[n] &= x_{2c}(nT) \\ &= \cos\left(2\pi \frac{10000n}{12000}\right) = \cos\left(\frac{5\pi n}{3}\right) \end{aligned}$$

$$\begin{aligned} \text{b) } &= \cos\left(\frac{5\pi n}{3} - 2\pi n\right) = \cos\left(\frac{(5-6)\pi n}{3}\right) \\ &\quad \uparrow \\ &\quad 2\pi n \text{ periodicity} \\ &= \cos\left(\frac{-\pi n}{3}\right) = \cos\left(\frac{\pi n}{3}\right) = x_1[n] \quad \text{QED} \\ &\quad \uparrow \\ &\quad \cos \text{ is even function} \end{aligned}$$

c) The same discrete-time sequence is consistent with two distinct continuous-time waveforms, so the samples alone cannot identify which input was present.